

VARUN S TERDAL

Gadag, Karnataka, India

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PROFESSIONAL SUMMARY

Computer Science (Artificial Intelligence) undergraduate passionate about solving real-world problems through AI research and software development. Experienced in building machine learning and Retrieval-Augmented Generation (RAG) systems, with a strong interest in Generative AI, information retrieval, and scalable intelligent applications. Continuously learning and applying emerging AI technologies through research and hands-on projects.

EDUCATION

Bachelor of Engineering in Computer Science (Artificial Intelligence)	2023 - 2027
KLE Technological University, Hubballi	CGPA: 7.45/10

TECHNICAL SKILLS

Programming	C++, Python, SQL
Core CS	DSA, OOP, DBMS, Operating Systems, Computer Networks
AI & ML	Machine Learning, Deep Learning, Generative AI, RAG, Federated Learning
Frameworks	PyTorch, TensorFlow, Scikit-learn, LlamaIndex
Developer Tools	Git, GitHub, MySQL

EXPERIENCE

Machine Thinking Intern	Jul 2025 – Sep 2025
KNIT Space, Hubballi	
- Engineered a cognitive AI search engine integrating Information Retrieval (IR), bias-aware reasoning, and game theory. - Built scalable preprocessing, indexing, and ranking pipelines for 1,500+ documents. - Enhanced retrieval quality through bias-aware ranking and optimized search workflows. - Improved search relevance by approximately 30% and reduced average query response time by 25%.	

PROJECTS

WiDS Datathon 2025 – ADHD Prediction
- Developed an end-to-end machine learning pipeline for early ADHD detection in women. - Achieved 87% classification accuracy through feature engineering and model optimization. - Secured Global Rank 22 in the WiDS Datathon 2025.

Chest X-ray Denoising System

- Developed a DnCNN-based deep learning model for low-dose chest X-ray denoising.
- Achieved 35 dB PSNR using improved MSE and SSIM loss functions.
- Research published at IEEE ITHREECT Conference 2026.

LexRAG – Legal Document Question Answering using RAG

- Developed a Retrieval-Augmented Generation (RAG) pipeline using LlamaIndex, semantic chunking, transformer embeddings, and vector search for legal document question answering.
- Integrated LLM-based retrieval to generate context-aware responses over legal documents.
- Achieved approximately 92% retrieval accuracy across 10,000+ legal documents.

Agentic AI–Optimized Asynchronous Federated Learning

- Designed an Agentic AI–optimized asynchronous federated learning framework using Flower and KubeEdge.
- Reduced communication overhead by 40%.
- Improved latency by 30% while enhancing scalability across edge-cloud environments.

ACHIEVEMENTS

- Secured Global Rank 22 in the WiDS Datathon 2025.
- Research paper published at IEEE ITHREECT Conference 2026.
- Pick and Place Bot recognized at Prayog Varsh 2024, KLE Technological University.
- Recognized for developing a DTSI solution to improve stray dog safety and promote community welfare.

CERTIFICATIONS

- Jenkins Configuration for DevOps (2026)
- Juniper Networks Certified Associate (JNCIA)
- Understanding DevOps Tools (DevOps Mindset) (2026)